

mm-Wave Acoustics:Tackling a Fundamental Limitation of Acoustic Wave Technologies

Sheena Selvakumar
Department of Electrical and Computer Engineering
Texas A&M University
Email: shes123@tamu.edu

Abstract—This paper presents the challenges and recent advancements in millimeter-wave (mm-wave) acoustic wave technologies for next-generation communication systems. Conventional acoustic wave (AW) devices are fundamentally limited to frequencies below 8 GHz due to scaling constraints and reduced electromechanical coupling. This work explores a novel approach based on inhomogeneous piezoelectricity using ferroelectric heterostructures to overcome these limitations. The proposed framework enables programmable resonators and mm-wave acoustic integrated circuits (AWICs), demonstrating high-frequency operation with improved performance metrics.

Index Terms—mm-wave, acoustic wave devices, piezoelectricity, ferroelectrics, resonators, RF systems

I. INTRODUCTION

The rapid growth in wireless data traffic and emerging applications such as 5G, 6G, IoT, and autonomous systems demand higher frequency operation and improved spectral efficiency. Acoustic wave (AW) devices are widely used in RF front-ends for filtering and sensing due to their low loss and compact size. However, conventional AW technologies are limited to sub-8 GHz frequencies, restricting their applicability in mm-wave systems. Overcoming this limitation is critical for enabling next-generation communication and sensing platforms.

II. MODULATOR DESIGN AND IMPLEMENTATION

Traditional scaling of AW devices to higher frequencies requires reducing thickness, which leads to increased loss, reduced coupling, and poor power handling. Additionally, harmonic resonance modes suffer from coupling degradation proportional to $1/f^2$, imposing a fundamental limitation.

To address this, inhomogeneous piezoelectricity is introduced using multilayer ferroelectric heterostructures. By exploiting electrostriction and ferroelectric domain switching, spatially varying piezoelectric coefficients are achieved. This enables enhanced coupling at higher frequencies and supports programmable resonator operation.

A lumped-element model is used to design programmable resonators with multiple operating modes. These devices allow selective excitation of resonance modes while suppressing others, enabling reconfigurable frequency responses. Materials such as Sc-doped AlN (AlScN) provide high electromechanical coupling and tunability, making them suitable for mm-wave applications.

III. EXPERIMENTAL RESULTS AND SYSTEM DEMONSTRATION

Experimental demonstrations show successful implementation of programmable acoustic resonators with multiple operational states. The proposed approach enables operation beyond 10 GHz, with reported devices achieving frequencies up to 20 GHz and high figures of merit.

Measured impedance responses validate the ability to switch between modes and control resonance behavior. The integration of mm-wave AWICs with RF circuits demonstrates their potential for compact, high-performance front-end modules. Compared to conventional approaches, the proposed design significantly improves frequency scalability and functionality.

Devices implemented using Sc-doped AlN (AlScN) demonstrate improved material characteristics, including enhanced coupling coefficients and tunability. Experimental results show operation beyond 10 GHz, with reported devices reaching up to 20 GHz and achieving high figures of merit (FOM), defined as $Q \times K^2 \times f$. These results represent a significant improvement over traditional acoustic devices.

Furthermore, integration of mm-wave acoustic integrated circuits (AWICs) with RF front-end modules highlights their potential for compact and scalable system design. The demonstrated programmability enables dynamic reconfiguration of frequency responses, making these devices suitable for multi-band and adaptive communication systems.

IV. CONCLUSION

This work addresses a fundamental limitation in acoustic wave technologies by introducing inhomogeneous piezoelectricity through ferroelectric heterostructures. The proposed framework enables programmable resonators and mm-wave acoustic devices, extending operation into the mm-wave regime. These advancements provide a promising pathway for next-generation RF systems, including 5G+ and 6G applications.

This work presents a novel approach to overcoming the fundamental frequency limitations of acoustic wave devices by leveraging inhomogeneous piezoelectricity in ferroelectric heterostructures. By addressing the intrinsic coupling degradation in harmonic resonance modes, the proposed framework enables efficient operation in the mm-wave regime.

Experimental results validate the feasibility of programmable acoustic resonators with high-frequency operation, improved coupling, and enhanced performance metrics. The successful demonstration of mm-wave AWICs establishes a new paradigm for RF acoustic technologies, enabling reconfigurable, compact, and high-performance front-end solutions.

Overall, this approach provides a scalable pathway toward next-generation communication and sensing systems, including 5G+, 6G, and beyond, where high-frequency operation and system integration are critical.